11 vs 12 Speed in MTB: Efficiency, Geometry and the Marginal Cost of the Extra Cog

Carlos Eduardo Ravello Joo

BikeLab Studio · Trujillo, Peru

ORCID: 0009-0007-5631-7436

ABSTRACT

Going from 11 to 12 speeds is sold as an obvious upgrade: one more cog, more range, a better bike. The engineering question is different: what exactly does that extra cog buy, and what does it cost? The answer is inconvenient for the marketing.

The real difference between a modern 12-speed group (Shimano M6100, 10–51T) and its 11-speed predecessor (M5100, 11–51T) is not at the large end of the cassette —both reach 51T— but at the small end: the 12-speed adds a **10-tooth** cog. That cog is what extends the range toward the fast gears. But a 10T cog is *measurably and necessarily* less efficient than an 11T: with fewer teeth, each chain link must articulate through a larger angle, and that articulation is paid as heat in the pins. Under load, the 10T dissipates roughly **0.5 % more** of the power than the 11T.

To that physical penalty a financial one is added: the 12-speed chain is narrower, wears out sooner, and the full ecosystem (chains and cassettes over the service life) costs appreciably more. The conclusion is sober: 12-speed is justified if you *need the range* or the simplicity of a single chainring; it is not justified by efficiency or economy. Paying more power and more money for a bigger number is, in most cases, a poor purchase.

Keywords: *MTB drivetrain, 11-speed, 12-speed, chain efficiency, link articulation, marginal cost, 10T cog, Shimano Deore.*

1. The question, asked honestly

"More speeds is better" is one of those phrases that sound obvious until you ask for the numbers. A drivetrain is not better for having more cogs; it is better if it delivers the range you need, with the least loss and the least maintenance cost. Those three things —range, efficiency, cost— do not move together, and the jump from 11 to 12 speeds separates them in an instructive way.

It is also worth bounding the comparison. Here we pit two directly related groups from the same brand and tier —Shimano Deore M5100 (11-speed) and M6100 (12-speed)— because comparing groups from different makers mixes in design choices unrelated to the question of cog count. What changes between these two is not the philosophy: it is, almost exclusively, the extra cog.

2. Geometry: where the difference actually is

The most common mistake is to imagine that 12-speed "stretches" the cassette at the large end, giving easier climbing gears. It does not. Both cassettes top out at 51T. The difference is at the opposite end: the 12-speed replaces the smallest 11T cog with a 10T and reorders the progression.

Position	11-speed — M5100	12-speed — M6100	Equivalence
1 (smallest)	11T	10T	Not equivalent
2	13T	12T	Not equivalent
3	15T	14T	Not equivalent
4	18T	16T	Not equivalent
…	… → 51T	… → 51T	Same top

Shimano Deore M5100 / M6100 specifications.

The 10T cog is, then, the whole argument for 12-speed: it extends the gearing upward, allowing more top speed, or smaller and lighter chainrings at the same speed. It is a genuine *range* advantage. The question is what is given in return.

3. Why a small cog loses more: the physics of articulation

A bicycle chain does not slide: it articulates. Every time a link enters and leaves a cog, the pin rotates inside its bushing under the full chain tension. That rotation, multiplied by hundreds of links per second, is the dominant source of frictional loss in the drivetrain.

The angle each link must articulate to wrap a cog is, geometrically, inversely proportional to the number of teeth:

$$\theta_{articulation} = 360° \div N_{teeth}$$

A 10-tooth cog forces each link to rotate 36°; an 11-tooth cog, 32.7°; a 21-tooth cog, just 17°. More rotation under load means more frictional work lost as heat at every pin. That is why chain efficiency falls systematically on small cogs, an effect measured repeatedly on the test bench:

Cog	Measured efficiency	Loss at 200 W
52T	99.2 %	1.6 W
21T	98.4 %	3.2 W
11T	97.1 %	5.8 W

Values consistent with the drivetrain-efficiency literature [1][2][3].

The reading is counterintuitive but firm: the *smallest* cog is always the *least* efficient. And 12-speed exists precisely to add one even smaller than the 11-speed's.

4. The concrete penalty of the 10T versus the 11T

If the small cog is the least efficient, comparing the 12-speed's 10T with the 11-speed's 11T isolates the exact physical cost of the "extra cog". At equal delivered power, the 10T dissipates more:

Power	Loss 11T → 10T	Extra loss	% of total power
150 W	2.9 W → 3.7 W	+0.8 W	0.53 %
200 W	3.9 W → 5.0 W	+1.1 W	0.55 %
300 W	5.9 W → 7.4 W	+1.5 W	0.50 %

Articulation-loss model at equal chain and lubrication.

The number to remember is **half a percentage point**. Riding the 10T cog instead of the 11T, about 0.5 % of your power goes out as extra heat —close to 1 W at 200 W. It is not a dramatic figure, and most riders will not feel it. But it points the wrong way: the "premium" component, the one that justifies the more expensive group, is the *least* efficient in the drivetrain. You pay more for the gear that performs worst.

5. Marginal cost: what the label does not say

Efficiency is only half the account. The other half is money, and here the difference is larger and easier to feel. The 12-speed chain is narrower than the 11-speed one to fit the extra cog in the same hub width; a narrower chain has less material in its plates and bushings, wears out sooner, and demands more frequent replacement. Over the drivetrain's service life, that adds up.

Item	11-speed	12-speed	Difference
Initial investment (group)	$145	$190	+$45
Chains over service life (≈7.5 vs 10)	$225	$350	+$125
Cassettes (≈2.5)	$163	$238	+$75
Estimated total difference			**≈ +$245**

2025–2026 reference prices, Deore tier; vary by region and brand.

12-speed does not just cost more to buy: it costs more to maintain, and most of the premium is not in the initial purchase but in the consumables you will replace over years. It is a decision with a long tail.

6. When 12-speed is the right call

None of the above says 12-speed is bad. It says you choose it for the right reason, not for the number. 12-speed is justified when:

- **You need the range and the simplicity of a single chainring.** Coming from a 2× or 3× and wanting one front ring without losing useful gears, 12-speed delivers that range with less mechanical complexity. Here it clearly wins.

- **You run out of gearing at the top.** The rider who genuinely "spins out" on fast sections uses the 10T; for them that cog is function, not luxury.
- **You want smaller, lighter chainrings** while keeping top speed, a secondary geometric advantage of the 10T.

Outside those cases, 11-speed delivers practically the same useful range, with better efficiency in its fastest gear and a lower running cost.

7. Conclusions

The jump from 11 to 12 speeds buys **range** and the convenience of a single chainring. It does not buy efficiency —its signature gear, the 10T, loses ~0.5 % more power than the 11T— nor economy —it costs on the order of $245 more over the service life. These are three variables the marketing fuses into one larger number, and that engineering separates.

For the rider who needs the range, 12-speed is the right tool and the premium is justified. For the one who does not —which is many— 11-speed is the rational choice: just as capable in practice, more efficient where it shows most, and cheaper to maintain. The number on the shifter is not a measure of quality.

8. Limitations

The cited efficiencies come from chain-friction tests under controlled conditions (isolated chain, uniform lubrication); the real system adds losses from the derailleur, chain cross-over and state of cleanliness, which can exceed the 10T–11T difference discussed here. The cost model uses mid-tier reference prices and typical service lives; both vary with region, brand and use. The comparison is limited to one pair of directly related groups (Deore M5100/M6100) to isolate the "cog count" variable; other tiers may shift the absolute figures without altering the qualitative conclusion, which is robust: the small cog is the least efficient, and 12-speed costs more to run.

References

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